

## 6.6 Prototype Development Cycle (Design, Build, Test, Iterate)

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔

CS XII

Chapter 6: Entrepreneurship in the Digital Age

Lecture 07 of 16

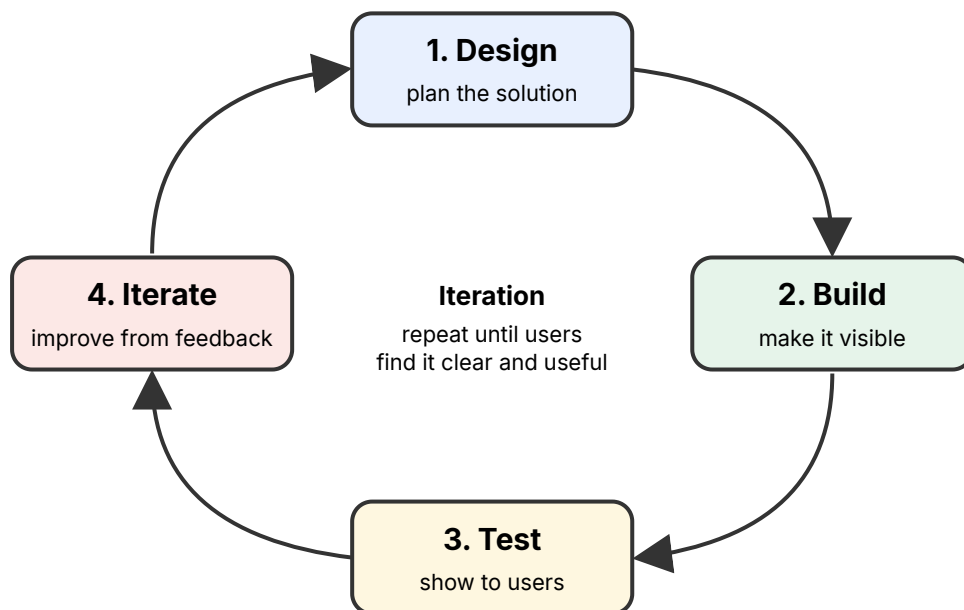
★ Golden

## 6.6 Prototype Development Cycle (Design, Build, Test, Iterate)

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### LEARN IT

Prototype development is **not a one-time activity**. A prototype is **created, tested, improved, and tested again**. This repeating process is called **iteration**. The main stages are **Design, Build, Test, and Iterate**.



The prototype development cycle: Design → Build → Test → Iterate, then back to Design.

| Stage     | What happens                                                                                                      |
|-----------|-------------------------------------------------------------------------------------------------------------------|
| 1. Design | Understand the problem and plan the solution. Decide what the prototype should show (e.g. screen flow, features). |
| 2. Build  | Create the first version (physical or digital). The aim is not beauty; the aim is to make the idea visible.       |

|                   |                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------|
| <b>3. Test</b>    | Show the prototype to users. Observe what they understand and what they find difficult.          |
| <b>4. Iterate</b> | Improve the prototype based on user feedback. Remove unnecessary parts and add missing features. |

#### WORKED EXAMPLE

**Worked example: two rounds of the cycle.** A team in Thatta wants a WhatsApp catalogue for a local Ajrak shop.

##### Round 1

- **Design:** The problem is that customers do not know which designs and prices are available. Plan: one page per design with photo, price and an "Order" message.
- **Build:** They make a paper booklet with 6 hand-drawn "catalogue pages".
- **Test:** They show it to 5 customers. Three ask, "Is this single or double-sided Ajrak? What size?"
- **Iterate:** They add size and type to every page.

##### Round 2

- **Design:** Add the size and type fields, and a delivery note.
- **Build:** They make the catalogue in WhatsApp Business with real photos.
- **Test:** 5 new customers browse it. Now nobody asks about size; one says the delivery charge is unclear.
- **Iterate:** They add "Delivery in Thatta: Rs 100" to each item.

Each round produces a clearer version, because each change comes from real feedback.

#### REMEMBER IT

**Mnemonic: "Do Big Things Iteratively"** = Design, Build, Test, Iterate. And remember the arrow at the end goes **back to Design**: it is a **cycle**, not a line.

#### COMMON MISTAKE

**Mistake 1:** Drawing the stages as a straight line that ends at "Iterate". **Fix:** Always draw a loop with an arrow from Iterate back to Design.

**Mistake 2:** Making the Build stage beautiful. **Fix:** The source says "The aim is not beauty; the aim is to make the idea visible."

پروٹوٹائپ ایک ہی بار میں نہیں بنتا؛ اسے بنایا، آزمایا، بہتر کیا اور دوبارہ آزمایا جاتا ہے۔ اس بار بار دہرانے کے عمل کو **Iteration** (تکرار) کہتے ہیں۔ چار مراحل ہیں: ڈیزائن (مسئلہ سمجھ کر حل کی منصوبہ بندی)، بلڈ (پہلا نمونہ بنانا)، ٹیسٹ (صارفین کو دکھانا) اور **Iteration** (رائے کی بنیاد پر بہتری)۔

#### EXAM TIP

A favourite long question: "Explain the prototype development cycle with a diagram." Draw the four boxes in a **circle** with arrows, write one or two lines for each stage using the source's words, define **iteration**, and finish with a short example (e.g. the ClassBuddy activity in 6.7). Marks are often lost for drawing a straight line instead of a cycle.

#### CHECK YOURSELF

1. What is iteration?
2. At which stage do you observe what users find difficult?
3. What is the aim of the Build stage?

Answers: 1. The repeating process of creating, testing, improving and testing a prototype again. 2. The Test stage. 3. To make the idea visible (not to make it beautiful).

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** The cycle is:

- a. Design → build → test → iterate (repeat)
- b. Build once, never look
- c. Test on no one
- d. Advertise first

**MCQ 2.** Iterate means:

- a. Change the prototype from what you learned
- b. Print more logos
- c. Hire 50 people
- d. Write OSI parsers

**MCQ 3.** This is ★ because:

- a. Entrepreneurship is not one-shot in the book
- b. Waterfall never iterates even in software
- c. Users are optional
- d. Ethics are optional

**MCQ 4.** If tests fail, you:

- a. Return to design
- b. Ship anyway always
- c. Delete the users
- d. Change the syllabus

**MCQ 5.** A cycle can be:

- a. Hours (paper) not months
- b. Only yearly
- c. Only after IPO
- d. Only in Silicon Valley

**MCQ 6.** Skipping test in the cycle turns it into:

- a. Just building in the dark
- b. HCI gold
- c. SQL gold
- d. Big O gold

## Extra short questions

1. Name the four stages.
2. What do you carry from test to design?
3. How many cycles is "enough" for class?
4. Link to Agile from XI.

## Extra long question

1. Document two cycles of a homework-reminder prototype. Each cycle: what you built, who tested, what changed. ★ Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Design, build, test, iterate.
2. Notes: where people stuck, what they asked for, what they ignored.
3. At least two documented rounds.
4. Same spirit: small loops with feedback. Don't over-claim they are identical.

### Long

1. Cycle1 paper: people wanted SMS not email. Cycle2 Figma with SMS mock. Metric: 4/5 could complete. Table.

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Next lecture: 6.7 Class Activity: Create a Prototype